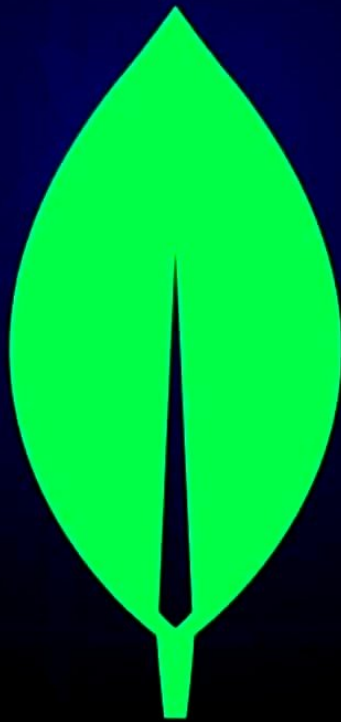


Node js

Notes



=> TechTalks_with_Ajay
@AjayTripathi



Intro



- Node js is a **server-side** JavaScript runtime environment.
- It is built on **Chrome's V8 engine**.
- It allows developers to use JavaScript to create **scalable**, **fast**, and **efficient** applications.
- its primary use is in **backend development**

CMD Prompts



```
1 // touch "filename.format" --> creates new file
2 // echo "text" --> print text on terminal
3 // mkdir "name" --> make directory
4 // cd "directory" --> to change directory
5 // cd .. --> go back to previous directory
6 // ls --> view files in directory
7 // ls -r --> view files in reverse order
8 // cp -r "source" "destination" --> copies files recursively from one place to another
9 // mv -r "source" "destination" --> same as copy but removes from original position
10
11 // node "filename" --> turn ur terminal to a browser console version of node js
12 //process.argv : Array of command-line arguments, including the Node.js executable path and script path.
13 //Example: `node script.js arg1 arg2` will set `process.argv` to `['node', 'script.js', 'arg1', 'arg2']`.
14 //process.version`*: Displays the installed Node.js version.
15 //process.cwd()`*: Returns the current working directory.
16 //process.release`*: Displays Node.js release information.
17
```

NPM RPEL

- **Read-Evaluate-Print Loop.**
- An **interactive shell** and **run time** environment for running JavaScript code.
- Similar to the browser's ``console.log()``.
- Cannot manipulate **DOM elements**.
- Provides a global object for immediate access to Node.js features.

Common js(require)

- Uses **require()** to import modules.
- Exports data or functions using **module.exports**.
- Loading is **synchronous**, making it simple but potentially slower for large modules.

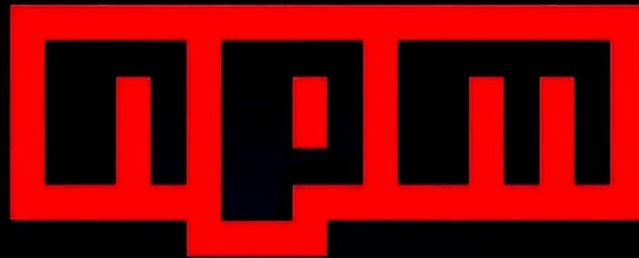
```
// 3. CommonJS Module System Example
const path = require("path"); // Built-in module
const add = (a, b) => a + b;
module.exports = { add };

console.log("CommonJS Example: Sum:", add(5, 10));
```

ES6(import/export)

- Requires **type="module"** in `package.json` to use.
- Uses **import** and **export** statements.
- Loading is **asynchronous**, improving performance.

```
// 4. ES6 Module System Example (requires type="module" in package.json)
import fs from 'fs';
export function multiply(a, b) {
  return a * b;
}
console.log("ES6 Example: Multiply:", multiply(2, 3));
```



- It is both a **library** of JavaScript packages and a **command-line** tool for managing them.
- these packages are stored in **node modules** (directory)

```
// ### Key Commands:  
// npm install <package>: Installs a package locally.  
// npm install -g <package>: Installs a package globally  
// npm link: Links global packages for local use.  
// npm uninstall <package>: Removes an installed package.
```


Package.json

- **Metadata file** for the project.
- Used for reinstalling dependencies and sharing project information.
- Created using ``npm init``.

```
{  
  "name": "cars",  
  "version": "1.0.1",  
  "description": "Car descriptions",  
  "main": "index.js",  
  "scripts": {  
    "test": "echo \"Error: no test specified\" && exit 1"  
  },  
  "author": "Eshaal",  
  "license": "ISC",  
  "type": "module",  
  "dependencies": {  
    "figlet": "^1.8.0"  
  }  
}
```



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